1)

To write an Assembly Language Program to get a string from the user and display it by writing values to ports 60h and 64h.

PROGRAM:

```
TITLE    PORTTXTIN.ASM -- Program to get input and display using port 60h/64h

;//**********************************************************
*********

;//*                                              *

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;//*                                              *

;//**********************************************************
*********

;<AMI_FHDR_START>

;

; Name: <port60.asm>

;

; Description:

; <Program to get user input and display using port 60h/64h>

;

;<AMI_FHDR_END>

;************************************************************
**********

.model small

.stack

;-----------------------------------------------------------------------
```

```
data segment
MSG1 DB 13,10,'Enter text: $'
MSG2 DB 13,10,'Output: $'
BUF  DB 20      ; max length
    DB ?        ; actual length
    DB 20 DUP(?) ; data
data ends
;--------------------------------------------------------------------------
code segment
assume cs:code, ds:data
start:
       MOV AX, data
       MOV DS, AX
       ; print input message
       LEA DX, MSG1
       MOV AH, 09H
       INT 21H
       ; read string
       LEA DX, BUF
       MOV AH, 0AH
       INT 21H
       ; print output label
       LEA DX, MSG2
       MOV AH, 09H
       INT 21H
       ; SI -> input string
```

```
    LEA SI, BUF+2
    MOV CL, BUF+1
    MOV CH, 0
NEXTCHAR:
    CMP CX, 0
    JE EXIT1
    ; write to port 64h
    MOV DX, 64H
    MOV AL, 0D2H
    OUT DX, AL
    ; write char to port 60h
    MOV DX, 60H
    MOV AL, [SI]
    OUT DX, AL
    ; display char
    MOV DL, [SI]
    MOV AH, 02H
    INT 21H
    INC SI
    LOOP NEXTCHAR
EXIT1:
    MOV AH, 4CH
    INT 21H
code ends
end start
;-----------------------------------------------------------------------------
```

OUTPUT:

```
C:\>masm port60.asm;
Microsoft (R) MASM Compatibility Driver
Copyright (C) Microsoft Corp 1993.  All rights reserved.

 Invoking: ML.EXE /I. /Zm /c /Ta port60.asm

Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993.  All rights reserved.

 Assembling: port60.asm

C:\>link port60.obj;

Microsoft (R) Segmented Executable Linker  Version 5.31.009 Jul 13 1992
Copyright (C) Microsoft Corp 1984-1992.  All rights reserved.


C:\>port60

Enter text: hello
Output: hello
C:\>_
```

2)

To write an Assembly Language Program to check CMOS checksum and display result after user interaction.

ALGORITHM:

- Start the program.
- Display message to user.
- Wait for key press.
- Read CMOS values and calculate checksum.
- Read stored checksum.
- Compare both values.
- Display result.
- Stop program.

PROGRAM:

```
TITLE    CMOSCHKIN.ASM -- Program to check CMOS checksum with user interaction

;//***************************************************************
*********

;//*                                                    *
```

```
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;//*                                                                     *
;//*****************************************************************
*********
;<AMI_FHDR_START>
;
; Name: <CMOS.asm>
;
; Description:
; <Program to check CMOS checksum after user interaction>
;
;<AMI_FHDR_END>
;*****************************************************************
**********

.model small
.stack

;-------------------------------------------------------------------------
data segment
MSG1 DB 13,10,'Press any key to check CMOS checksum...$'
MSG2 DB 13,10,'CMOS checksum is proper.$'
MSG3 DB 13,10,'CMOS checksum is not proper.$'

SUM1 DW 0000H
```

```
SUM2 DW 0000H
data ends
;---------------------------------------------------------------------------

code segment
assume cs:code, ds:data

start:
    MOV AX, data
    MOV DS, AX

    ; print prompt message
    LEA DX, MSG1
    MOV AH, 09H
    INT 21H

    ; wait for key press
    MOV AH, 01H
    INT 21H

    ; initialize checksum sum
    MOV SI, 10H
    MOV SUM1, 0000H

READ_LOOP:
    CMP SI, 2EH
```

```
    JAE READ_STORED

    ; select CMOS offset
    MOV DX, 70H
    MOV AX, SI
    OUT DX, AL

    ; read CMOS data
    MOV DX, 71H
    IN  AL, DX

    XOR AH, AH
    ADD SUM1, AX

    INC SI
    JMP READ_LOOP

READ_STORED:
    ; read stored checksum high byte from 2Eh
    MOV DX, 70H
    MOV AL, 2EH
    OUT DX, AL

    MOV DX, 71H
    IN  AL, DX
    MOV AH, AL
```

```
        ; read stored checksum low byte from 2Fh
        MOV DX, 70H
        MOV AL, 2FH
        OUT DX, AL

        MOV DX, 71H
        IN  AL, DX

        MOV SUM2, AX

        ; compare calculated and stored checksum
        MOV AX, SUM1
        CMP AX, SUM2
        JE PROPER

NOT_PROPER:
        LEA DX, MSG3
        MOV AH, 09H
        INT 21H
        JMP EXIT1

PROPER:
        LEA DX, MSG2
        MOV AH, 09H
        INT 21H
```

```
EXIT1:
	MOV AH, 4CH
	INT 21H


code ends
end start
;---------------------------------------------------------------------------
```

OUTPUT:

```
C:\>masm CMOS.asm
Microsoft (R) MASM Compatibility Driver
Copyright (C) Microsoft Corp 1993.  All rights reserved.

 Invoking: ML.EXE /I. /Zm /c /Ta CMOS.asm

Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993.  All rights reserved.

 Assembling: CMOS.asm

C:\>link CMOS.obj;

Microsoft (R) Segmented Executable Linker  Version 5.31.009 Jul 13 1992
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C:\>CMOS

Press any key to check CMOS checksum...m
CMOS checksum is not proper.
C:\>_
```